# 12th Hackers Ka Jugaad.

## Chemical Kinetics

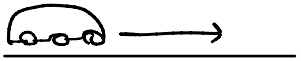
**Chemical Kinetics** → ① feasibility of a chemical reaction
↓
Thermodynamic
$\Delta G = +ve$ (non-spon.) $\Delta G = -ve$ (spon. reaⁿ)

② extent of chemical reaction: chemical equilibrium

③ speed of chemical reaction: Chemical Kinetics.

**Chemical Kinetics**
↓
rate of chemical reaction

Physics –



$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$\text{rate} = \frac{\text{change in amount of R/P}}{\text{time}}.$$

**The branch of chemistry, which deals with the study of reaction rates and their mechanisms, called chemical kinetics.**

Types of reaction

Very fast | Moderate | Very slow reaⁿ.

a) acid-base rea^n
b) precipitation rea^n

a) decomposition of $N_2O_5$
b) hydrolysis of starch
c) Inversion of can sugar

a) rusting of Fe

amount of reactant / Product.

- Moles
- Conc. of R/P
- Pressure

rate of rea^n = change in conc^n of reactant/Product / time

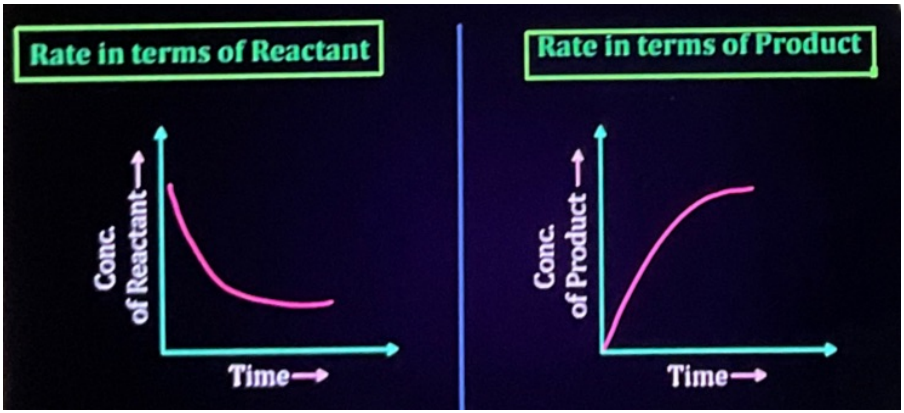
# Rate of chemical reaction

* The speed of a reaction or the rate of a reaction can be defined as the change in concentration of a reactant or product in unit time.

Rate can be expressed in terms of

(i) The rate of decrease in concentration of any one of the reactants, or

(ii) The rate of increase in concentration of any one of the products.



# rate of reaction -

$$A \longrightarrow 2B$$

| | | |
|---|---|---|
| $t=0$ | $a_0$ | 0 |
| $t=t$ | $(a_0-x)$ | $2x$ |

$a_t = a_0 - x$

Rate of reaction in terms of reactant $= \dfrac{\text{change in conc}^n \text{ of reactant}}{\text{time}}$

$$= \frac{C_f - C_i}{t_f - t_i} = \frac{a_0 - x - a_0}{t}$$

note: -ve represent dec. in conc$^n$ of reactant.

$$\text{Rate} = \frac{-x}{t} = -\left[\frac{-x}{t}\right]$$

Rate of reaction in terms of product $= \dfrac{C_f - C_i}{T_f - T_i} = \dfrac{2x-0}{t-0} = \dfrac{+2x}{t}$

Note: +ve represent inc. in conc. of product $\bar{c}$ time

ROR in terms of reactant $= -\dfrac{[\Delta C]}{[\Delta t]}$

ROR in terms of product $= +\dfrac{\Delta C}{\Delta t}$

## Individual rate of reaction

$$A + 2B \longrightarrow 3C$$

$2 \times R_A = R_B$

$R_A = \dfrac{R_B}{2}$ ---(1)

$R_A \neq R_B \neq R_C$

$3 \times R_A = R_C$

$R_A = \dfrac{R_C}{3}$ ---(2)

$$\frac{R_A}{1} = \frac{R_B}{2} = \frac{R_C}{3} = \text{ROR}$$

$$A + 2B \longrightarrow 3C$$

$$\text{ROR} = \frac{R_A}{1} = \frac{R_B}{2} = \frac{R_C}{3}$$

a) ROA / ROF = rate of form$^n$ of product.

b) ROD = rate of disappearance of

reactant.

$A + 2B \longrightarrow 3C$

1.) R.O.D of $A = -\frac{[\Delta A]}{\Delta t}$

2.) ROD of $B = -\frac{[\Delta B]}{\Delta t}$

3.) ROA/ROF $= \frac{+[\Delta C]}{\Delta t}$

$$ROR = -\frac{\Delta A}{\Delta t} = -\frac{1}{2}\frac{\Delta B}{\Delta t} = +\frac{1}{3}\frac{\Delta C}{\Delta t}$$

$aA + bB \longrightarrow cC + dD$ → Balanced Chemical Equation

ROD of $A = -\frac{\Delta[A]}{\Delta t}$

ROD of $B = -\frac{\Delta[B]}{\Delta t}$

ROF/ROA of $C = +\frac{\Delta[C]}{\Delta t}$

ROA of $D = +\frac{\Delta[D]}{\Delta t}$

$$ROR = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\frac{\Delta[B]}{\Delta t} = +\frac{1}{c}\frac{\Delta[C]}{\Delta t} = +\frac{1}{d}\frac{\Delta[D]}{\Delta t}$$

→ 'Inc in Conc. of Product'

Imp –

ROD/ROA/ROF ⇒ No need of Molar Coeff.

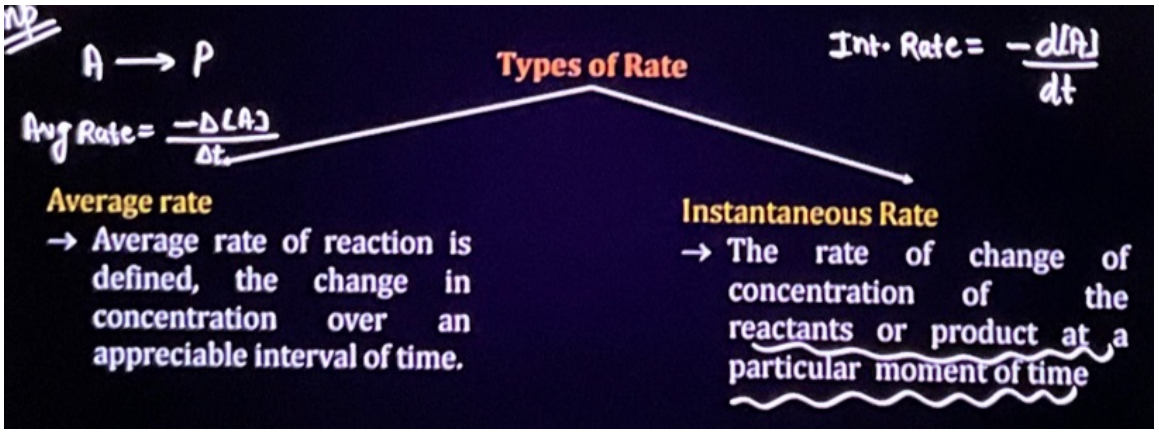
ROR : (Rate of Reaction)

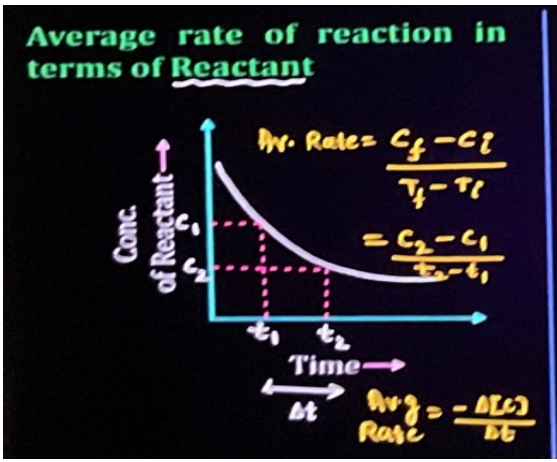
1) Balanced chemical Eq.

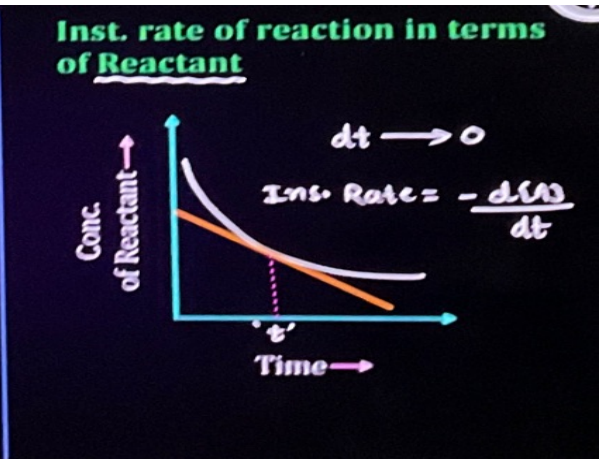
2) Molar Coeff include

**Average rate** – defined by macroscopic time interval.

## Types of Rate –




**Instantaneous rate** – defined by microscopic time interval.

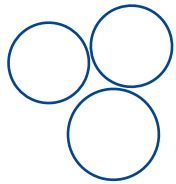
**units of rate of a reaction** – Rate = $\frac{\text{change in conc}^n}{\text{time}}$

$= \frac{\text{mol}}{\text{lit. sec}}$

conc. $= \frac{\text{mol}}{\text{lit.}}$

Unit of rate $= \text{mol lit}^{-1}\text{sec}^{-1}$

**Molecularity –** total no. of reacting species in an elementary reaction.



} total no. of colliding species

## Types of chemical reaction

**elementary chemical reaction**

— one-step chemical reaction

— $SN^2$ reaction

**Complex chemical reaction**

— multiple step-reaction

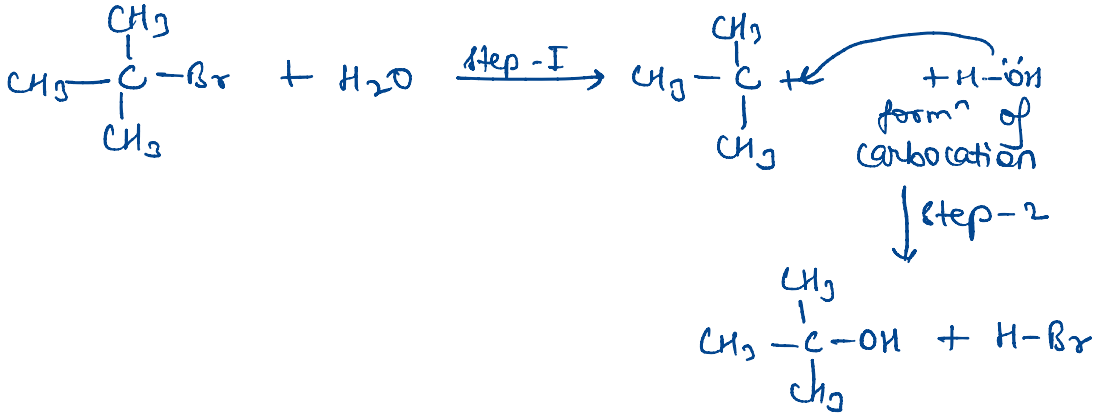
— eg = SN1 reaction.

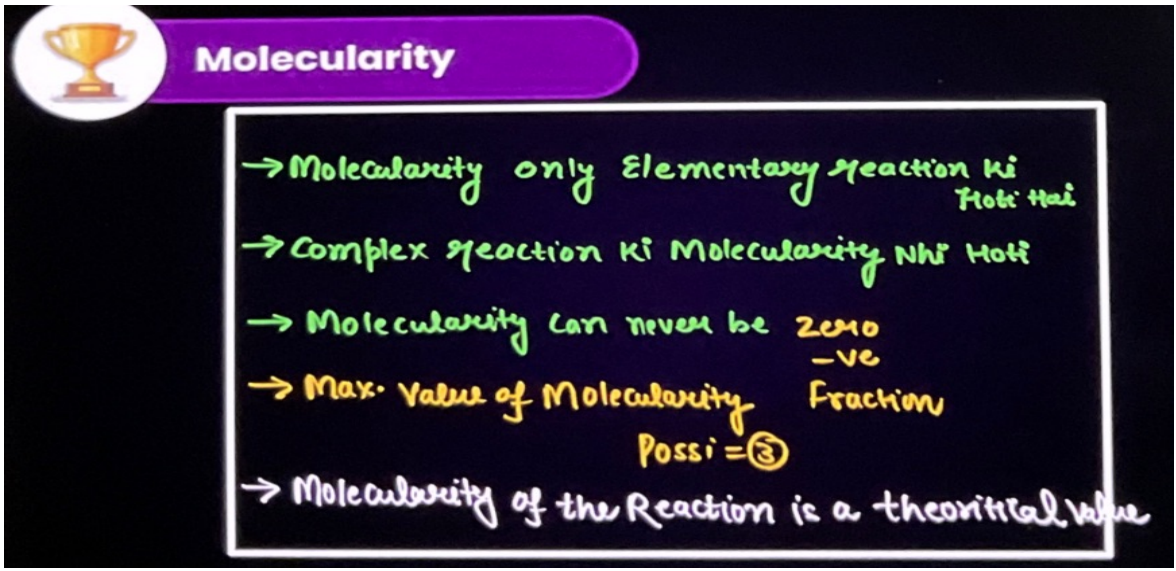
## Elementary reaction : one-step reaction

$$R-X + \bar{O}H \longrightarrow R-OH + X^-$$

this above rea^ is one step.

## Complex- rea^ –






## Order of reaction –

$xA + yB \longrightarrow zC$

→ sum of power of conc. of reactant in the rate law expression

$\text{Rate} \propto [R]^x$ — order of reaction

$\text{Rate} \propto [A]^m [B]^n$

$\text{Rate} = K[A]^m[B]^n$ — Rate law expression

Order of reaction $= m + n$

| | |
|---|---|
| $m = x$ | $n = y$ |
| $m \neq x$ | $n \neq y$. |

## Order –

In rate law equation sum of stoichiometric coefficients of concentration of reactants is called order of reaction. It is experiment value

# Reaction ko dekh kar order of Reaction Nhi Bta skte

1.) Order is for both elementary as well as complex reaction

2.) Expt. value

3.) Order of reaction
- → zero
- → fractional
- → +ve, -ve

$m = x$ $\quad n = y$
$m \neq x$ $\quad n \neq y$

Note: for an elementary reaction
Molecularity = order of reaction

**Order of reaction**

- zero order reac^n
- first order reac^n
- Pseudo first order reac^n.

**Unit of rate constant (k) for zero order reaction**

$\text{rate} = K[R]^0$

$K$ = rate constant.

$\frac{\text{conc.}}{\text{time}} = K$

$K_0 = \text{mol lit}^{-1}\text{sec}^{-1}$

$\frac{\text{mol}}{\text{lit} \times \text{sec}} = K_0$

**Unit of rate constant for 1st order reaction.**

$\text{Rate} = K[R]^1$

$\frac{\text{conc.}}{\text{time}} = K \times \text{conc.}$

$\frac{\text{conc}}{\text{time} \times \text{conc.}} = K$

$K = \text{sec}^{-1} / \text{time}^{-1}$

# unit of rate constant for 2nd order reaction

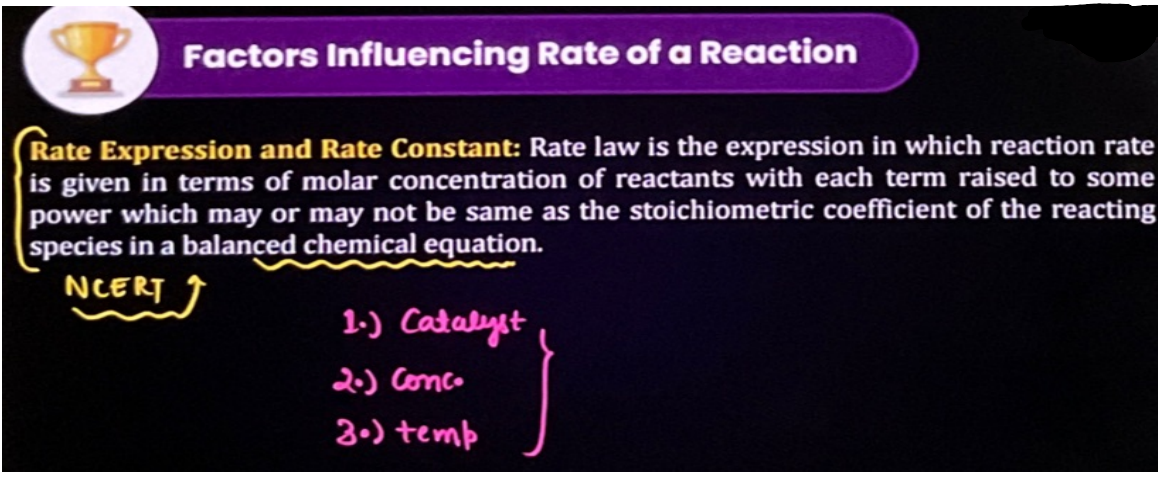
$$\text{Rate} = K[R]^2$$

$$\frac{\text{conc}}{\text{time}} = K(\text{conc})^2$$

$$\frac{\text{conc}}{\text{time} \times \text{conc}^2} = K$$

$$K = \text{conc}^{-1} \cdot \text{time}^{-1}$$

$$= \text{mol}^{-1}\ \text{lit.}\ \text{sec}^{-1}$$




# Zero order Reaction.

$$\text{Rate} = K[A]^0$$

$$\text{Rate} = K$$

note:

1.) for zero order rea$^n$, rate of rea$^n$ is independent of conc$^n$

2.) for zero order rate is always constant

$$A \longrightarrow P$$

| | | |
|---|---|---|
| $t=0$ | $A_0$ | 0 |
| $t=t$ | $(A_0 - x)$ | $x$ |

$$A_t = A_0 - x$$

$$\text{Rate} = \frac{-dA}{dt} = \frac{+dP}{dt} = K$$

$$\frac{-dA}{dt} = K$$

$$\int_{A_0}^{A_t} -dA = \int_0^t K \cdot dt$$

$$-[A]_{A_0}^{A_t} = K[t]_0^t$$

$$-[A_t - A_0] = Kt$$

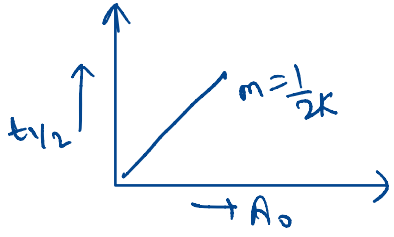
$$A_0 - A_t = Kt$$

$$\boxed{\frac{A_0 - A_t}{t} = K.}$$

$$\Rightarrow \frac{A_0 - A_0 + x}{t} = K$$

$$= \frac{x}{t} = K \longrightarrow \boxed{x = Kt}$$

## Half life for zero order reaction




$$A \longrightarrow P$$

| | | |
|---|---|---|
| $t=0$ | $A_0$ | 0 |
| $t=t$ | $(A_0 - x)$ | $x$ |

$$\boxed{A_0 - x = \frac{A_0}{2}} = t_{1/2}$$

$$x = \frac{A_0}{2}$$

$$K = \frac{x}{t}$$

$$t_{1/2} = \frac{x}{K}$$

$$\boxed{t_{1/2} = \frac{A_0}{2K}}$$

$y = mx$

$m = \frac{1}{2K}$ = slope

time of comp for zero order rea$^n$ $t_{comp}$.

$$x = A_0$$

$$t_{comp} = \frac{x}{K}$$

$$\boxed{t_{comp} = \frac{A_0}{K}}$$

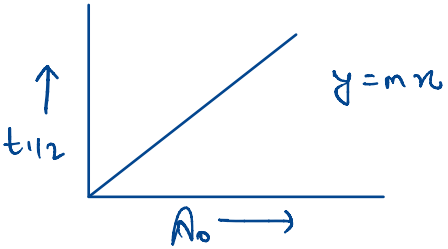
relation b/w $t_{1/2}$ and $t_{comp}$ for zero order rean.

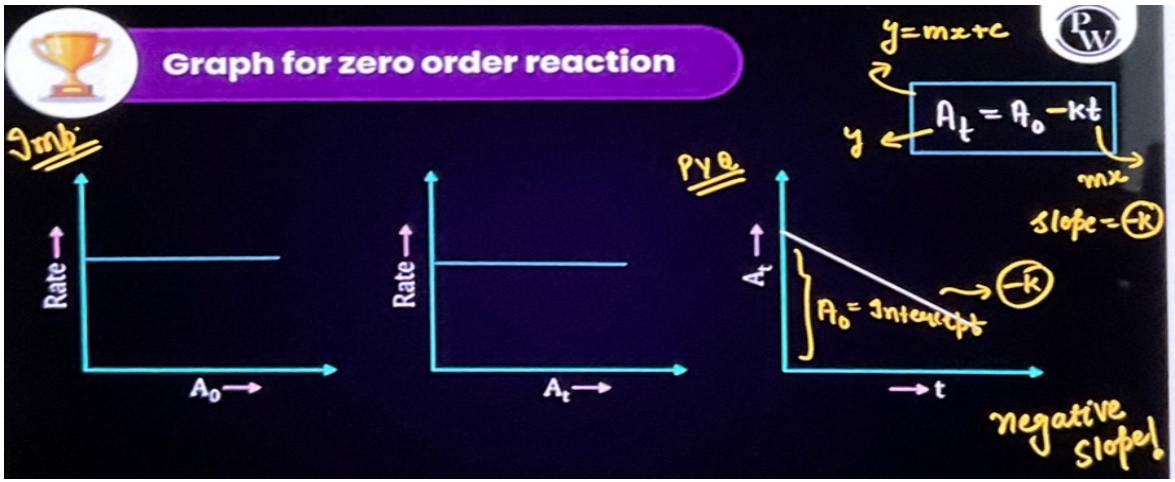
$$t_{1/2} = \frac{A_0}{2K} - t_{comp.}$$

$$t_{comp} = \frac{A_0}{K}$$

$$2 \times t_{1/2} = t_{comp.}$$

half life for zero order reaction




## First order reaction

$$A \longrightarrow P$$

| | A | P |
|---|---|---|
| t=0 | $A_0$ | 0 |
| t=t | $(A_0 - x)$ | x |

$$A_t = A_0 - x$$

$$\text{Rate} = K[A_t]^1$$

$$\frac{-dA}{dt} = K[A_t]^1$$

$$\int_{A_0}^{A_t} \frac{-dA}{A_t} = \int_0^t K \cdot dt$$

$$-\left[\ln A_t\right]_{A_0}^{A_t} = k\left[t\right]_0^t$$

$$-\left[\ln A_t - \ln A_0\right] = kt$$

$\ln A_0 - \ln A_t = kt$

$$\ln \frac{A_0}{A_t} = kt$$

$$2.303 \log \frac{A_0}{A_t} = kt$$

$\ln$ = natural log

log = common log

$\ln x = 2.303 \log x$

$A_t = A_0 - x$

$$K = \frac{2.303}{t} \log \frac{A_0}{A_0 - x}$$

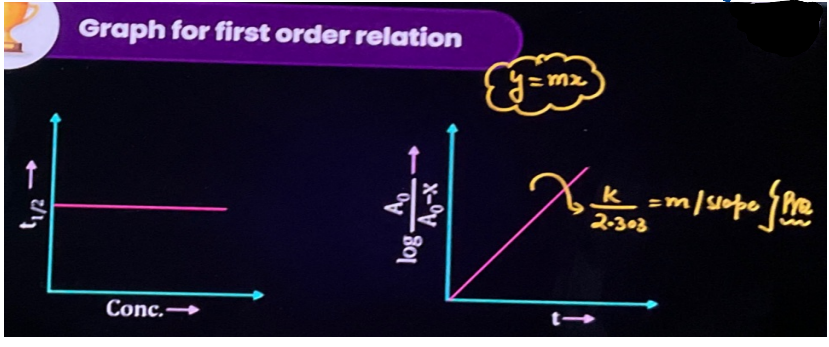
half life –

$$t_{1/2} = \frac{2.303}{K} \log \frac{A_0}{A_0/2}$$

$$= \frac{2.303}{K} \log 2$$

$$t_{1/2} = \frac{0.693}{K}$$

– half life = independent of conc$^n$.



## for gaseous phase: first order reaction.

$$A_{(g)} \longrightarrow B_{(g)} + C_{(g)}$$

| | | | |
|---|---|---|---|
| $P_0$ | 0 | 0 | $t=0$ |
| $(P_0 - P)$ | $P$ | $P$ | $t=t$ |

$$P_T = P_0 - P + P + P$$

$$P_T = P_0 + P$$

$$P_T - P_0 = P \quad \text{or} \quad \boxed{P = P_T - P_0}$$

from first order reaction:-

$$K = \frac{2.303}{t} \log \frac{A_0}{A_0 - x}$$

in terms of pressure -

$$K = \frac{2.303}{t} \log \frac{P_0}{P_0 - P}$$

$$K = \frac{2.303}{t} \log \frac{P_0}{P_0 - (P_T - P_0)}$$

$$K = \frac{2.303}{t} \log \frac{P_0}{P_0 - P_T + P_0}$$

$$K = \frac{2.303}{t} \log \frac{P_0}{2P_0 - P_T}$$

## Pseudo first order reaction.

The reaction which is of higher order reaction by exp. that reaction is first order reaction.

when conc. of one of the reactant is taken in large excess

$$CH_3COOC_2H_5 + H_2O \longrightarrow CH_3COOH + C_2H_5OH$$

(ethyl acetate) In large excess

$$\text{Rate} = K[H_2O][CH_3COOC_2H_5]$$

(const.) $k' = k[H_2O]$

$Rate = k[CH_3COOC_2H_5]$ — Pseudo 1st order reaction

## Temperature Dependence of the Rate of a Reaction

**Temperature Coefficient (μ):** Ratio of rate of reaction or rate constant of reaction at temperature differ by 10° C.

$$\mu = \frac{R_1}{R_2} = \frac{k_1}{k_2}$$ Imp

$k_1, k_2$ = Rate constant

$R$ = Rate of the reaction

Temp → R

T + 10°C → 'double'

For a chemical reaction with rise in temperature by 10°C, the rate constant is nearly doubled.

$$\mu = \frac{R_1}{R_2} = \frac{k_1}{k}$$ Imp

## Arrhenius equation

→ Temp. dependance of rate of reaction

$$k = A e^{-E_a/RT}$$

$k$ — rate const.; $A$ — freq. factor or pre exponential factor; $E_a$ — activation energy; $T$ — temp.; $R$ — gas const.

$A \rightarrow B$

at temp $T_1$ $\quad k_1 = A e^{-E_a/RT_1}$

at temp $T_2$ $\quad k_2 = A e^{-E_a/RT_2}$

$k_2 \div k_1$

$$\frac{k_2}{k_1} = e^{-E_a/RT_2 + E_a/RT_1} = e^{E_a/R\left[\frac{1}{T_1} - \frac{1}{T_2}\right]}$$

taking ln on both side

$$\ln \frac{K_2}{K_1} = \ln e^{E_a/R\left(\frac{1}{T_1}-\frac{1}{T_2}\right)}$$

$$\ln \frac{K_2}{K_1} = \frac{E_a}{R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right) \ln e^{1}$$

$$2.303 \log \frac{K_2}{K_1} = \frac{E_a}{R}\left[\frac{1}{T_1}-\frac{1}{T_2}\right]$$

$K = Ae^{-E_a/RT}$

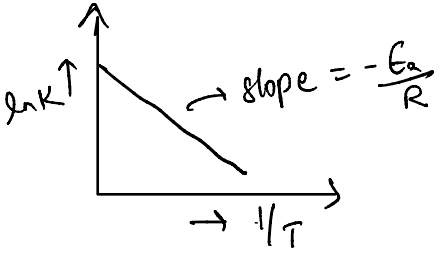
taking ln Both side

$$\ln K = \ln A e^{-E_a/RT}$$

$$\ln K = \ln A + \frac{-E_a}{RT} \ln e$$

$$\ln K = \ln A - \frac{E_a}{RT} \ln e$$

$$\boxed{\ln K = \ln A - \frac{E_a}{RT}}$$




## Transition state theory.

1. According to this, during a chemical reaction, there will be collision between different reactant molecules.
2. During these collisions, the existing bonds of reactant molecules will be broken and new bonds of product molecules will be formed.
3. All random collisions will not lead into the product formation.

4. Collisions which are taking place in proper orientation and with sufficient amount of energy ($\geq E_a$) will lead into the product formation.

$\geq E_a$

$E_a$ = Activation Energy

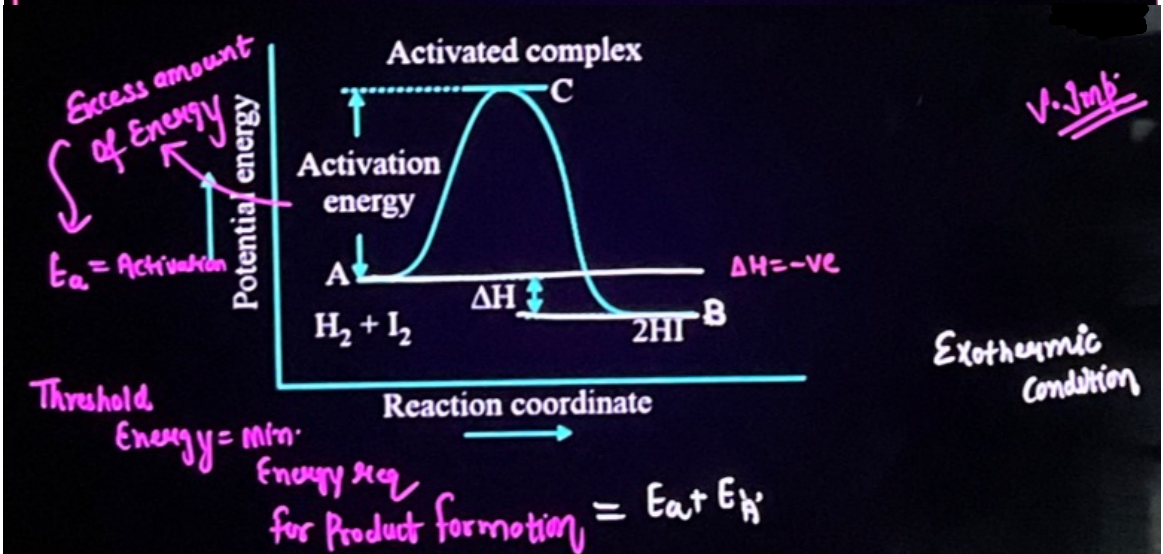
Excess amount of Energy given to the Reactant Molecule for the Formation of Product

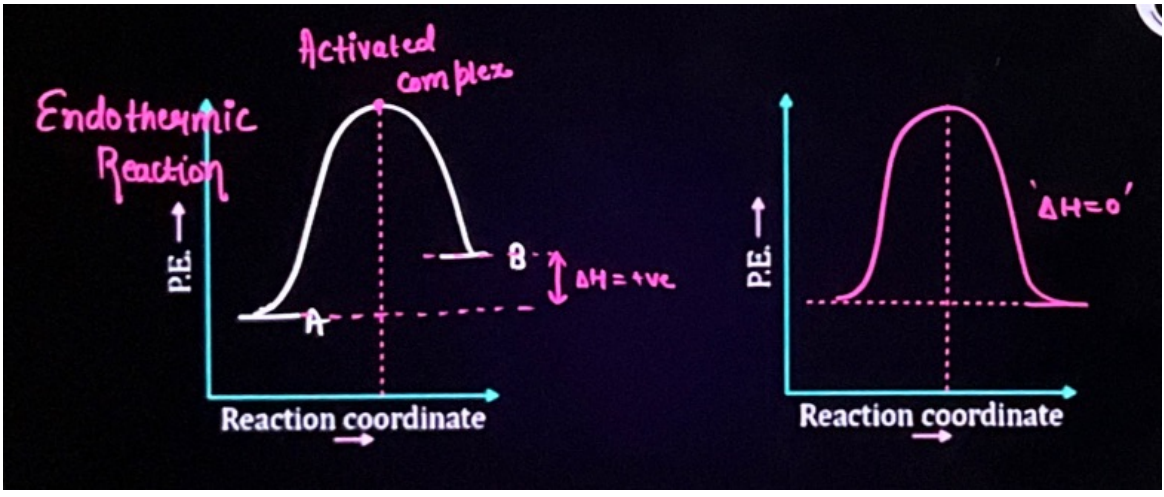
5. During effective collisions, there can be considered formation of a hypothetical species known as activated complex.

'Short' Imp.

6. In activated complex reactant bonds are not completely broken and products bonds are not yet completely formed.

- **Transition state:** Very unstable due to high energy.
- **Activation Energy:** The minimum amount of energy required to make active participate (transition state).
- **Threshold Energy:** Minimum amount of energy which the colliding molecules must possess in order to make the chemical reaction to occur.






## Effect of Catalyst –

A catalyst is a substance which increases the rate of reaction without itself undergoing any permanent chemical change

Catalyst provide alternate pathway of lower activation, energy

1.The substance which reduces the rate of reaction is called inhibitor.

2.The action of catalyst can be explained by intermediate complex theory.

3.According to this theory, a catalyst participates in a chemical reaction by forming temporary bonds with the reactants resulting in an intermediate complex. This has a transitory existence and decomposes to yield products and the catalyst.

4.The catalyst provides an alternate pathway of reaction mechanism by reducing activation energy between the reactants and products and hence lowering the potential energy Barrier